NAME: ABDUL GHAFFAR

REGISTRATION NO: BF25NWELE0711

DEPARTMENT: ELECTRICAL ENGINEERING (POWER)

SECTION: A
SUBJECT: CPP

ASSIGNMENT NO#: 02

SUBMITTED TO: SIR PROFESSOR RIZWAN .

TASK 01 : Find Maximum, Minimum, And Average.

```
#include <iostream>

using namespace std;


int main() {

   int arr[10];

   int max, min;

   float sum = 0;


   cout << "Enter 10 integers:\n";

   for(int i=0;i<10;i++) {

      cin >> arr[i];

   }


   max = min = arr[0];


   for(int i=0;i<10;i++) {

      if(arr[i] > max) max = arr[i];

      if(arr[i] < min) min = arr[i];

      sum += arr[i];

   }

   cout << "Maximum = " << max << endl;

   cout << "Minimum = " << min << endl;

   cout << "Average = " << sum/10 << endl;
```

```
    return 0;
}
```

TASK 02 : REVERSE A 1D ARRAY USING FUNTIONS.

```
#include <iostream>
using namespace std;

void reverseArray(int arr[], int size) {
    for(int i=0;i<size/2;i++) {
        int temp = arr[i];
        arr[i] = arr[size-i-1];
        arr[size-i-1] = temp;
    }
}

int main() {
    int arr[5];

    cout << "Enter 5 elements:\n";
    for(int i=0;i<5;i++) cin >> arr[i];

    cout << "Before Reverse:\n";
    for(int i=0;i<5;i++) cout << arr[i] << " ";

    reverseArray(arr,5);
```

```
    cout << "\nAfter Reverse:\n";
    for(int i=0;i<5;i++) cout << arr[i] << " ";

    return 0;
}
```

TASK 03: MATRIX ADDITION.

```
#include <iostream>
using namespace std;

int main() {
    int A[3][3], B[3][3], C[3][3];

    cout << "Enter Matrix A:\n";
    for(int i=0;i<3;i++)
        for(int j=0;j<3;j++)
            cin >> A[i][j];

    cout << "Enter Matrix B:\n";
    for(int i=0;i<3;i++)
        for(int j=0;j<3;j++)
            cin >> B[i][j];

    for(int i=0;i<3;i++)
        for(int j=0;j<3;j++)
            C[i][j] = A[i][j] + B[i][j];
```

```
    cout << "Result Matrix:\n";
    for(int i=0;i<3;i++) {
        for(int j=0;j<3;j++)
            cout << C[i][j] << " ";
        cout << endl;
    }

    return 0;
}
```

TASK 04: SORT NAMES ALPHABATICALLY.

```
#include <iostream>
using namespace std;

int main() {
    string names[5];

    cout << "Enter 5 names:\n";
    for(int i=0;i<5;i++) cin >> names[i];

    for(int i=0;i<5;i++) {
        for(int j=i+1;j<5;j++) {
            if(names[i] > names[j]) {
                string temp = names[i];
                names[i] = names[j];
```

```
            names[j] = temp;
          }
        }
    }

    cout << "Sorted Names:\n";
    for(int i=0;i<5;i++) cout << names[i] << endl;

    return 0;
}
```

TASK 05 : POWER SYSTEM LOAD ANALYSIS USING FUNTIONS.

```
#include <iostream>
using namespace std;

float totalLoad(float arr[]) {
    float sum = 0;
    for(int i=0;i<24;i++) sum += arr[i];
    return sum;
}

float peakLoad(float arr[]) {
    float max = arr[0];
    for(int i=1;i<24;i++)
        if(arr[i] > max) max = arr[i];
    return max;
```

```
}

int main() {
    float load[24];

    cout << "Enter load for 24 hours:\n";
    for(int i=0;i<24;i++) cin >> load[i];

    cout << "Total Load = " << totalLoad(load) << endl;
    cout << "Peak Load = " << peakLoad(load) << endl;

    return 0;
}
```

TASK 06: STUDENTS RECORD SYSTEM USING FUNTIONS.

```
#include <iostream>
using namespace std;

int findTopper(int marks[], int size) {
    int maxIndex = 0;
    for(int i=1;i<size;i++)
        if(marks[i] > marks[maxIndex]) maxIndex = i;
    return maxIndex;
}

void passFail(int marks[], int size) {
```

```
    for(int i=0;i<size;i++) {
        if(marks[i] >= 50)
            cout << "Pass\n";
        else
            cout << "Fail\n";
    }
}

int searchStudent(string names[], int size, string key) {
    for(int i=0;i<size;i++)
        if(names[i] == key) return i;
    return -1;
}

int main() {
    string names[5];
    int marks[5];

    cout << "Enter names and marks:\n";
    for(int i=0;i<5;i++)
        cin >> names[i] >> marks[i];

    int topper = findTopper(marks,5);
    cout << "Topper = " << names[topper] << endl;
```

```
    passFail(marks,5);

    string key;
    cout << "Enter name to search: ";
    cin >> key;

    int index = searchStudent(names,5,key);
    if(index != -1)
        cout << "Marks = " << marks[index];
    else
        cout << "Student not found";

    return 0;
}
```

TASK 07: RENEWABLE ENERGY MONITORING USING FUNTIONS.

```
#include <iostream>
using namespace std;

float totalEnergy(float arr[]) {
    float sum = 0;
    for(int i=0;i<24;i++) sum += arr[i];
    return sum;
}

int maxHour(float arr[]) {
```

```
    int index = 0;
    for(int i=1;i<24;i++)
        if(arr[i] > arr[index]) index = i;
    return index;
}

float avgEnergy(float arr[]) {
    return totalEnergy(arr)/24;
}

int main() {
    float energy[24];

    cout << "Enter energy for 24 hours:\n";
    for(int i=0;i<24;i++) cin >> energy[i];

    cout << "Total Energy = " << totalEnergy(energy) << endl;
    cout << "Hour with Maximum Generation = " << maxHour(energy) << endl;
    cout << "Average Energy = " << avgEnergy(energy) << endl;

    return 0;
}
```

TASK 08 : BATTERY HEALTH MONITORING USING FUNTIONS.

```
#include <iostream>
```

```
using namespace std;

int countBelow(float arr[]) {
  int count = 0;
  for(int i=0;i<20;i++)
    if(arr[i] < 3.0) count++;
  return count;
}

int countAbove(float arr[]) {
  int count = 0;
  for(int i=0;i<20;i++)
    if(arr[i] > 4.2) count++;
  return count;
}

float findMin(float arr[]) {
  float min = arr[0];
  for(int i=1;i<20;i++)
    if(arr[i] < min) min = arr[i];
  return min;
}

float findMax(float arr[]) {
  float max = arr[0];
```

```
    for(int i=1;i<20;i++)
        if(arr[i] > max) max = arr[i];
    return max;
}

int main() {
    float voltage[20];

    cout << "Enter 20 voltage readings:\n";
    for(int i=0;i<20;i++) cin >> voltage[i];

    int below = countBelow(voltage);
    int above = countAbove(voltage);

    cout << "Below Safe Limit = " << below << endl;
    cout << "Above Safe Limit = " << above << endl;
    cout << "Minimum Voltage = " << findMin(voltage) << endl;
    cout << "Maximum Voltage = " << findMax(voltage) << endl;

    if(below > 0 || above > 0)
        cout << "Warning: Unsafe readings exist!\n";

    return 0;
}
```

TASK 09: SMART GRID DISTRIBUTION USING FUNTIONS.

```
#include <iostream>

using namespace std;


void regionTotal(int arr[4][7]) {

    for(int i=0;i<4;i++) {

        int sum=0;

        for(int j=0;j<7;j++) sum += arr[i][j];

        cout << "Region " << i+1 << " Total = " << sum << endl;

    }

}


int highestDay(int arr[4][7]) {

    int maxDay=0, maxSum=0;


    for(int j=0;j<7;j++) {

        int sum=0;

        for(int i=0;i<4;i++) sum += arr[i][j];


        if(sum > maxSum) {

            maxSum = sum;

            maxDay = j;

        }

    }

    return maxDay;

}
```

```
float overallAverage(int arr[4][7]) {
    int total=0;
    for(int i=0;i<4;i++)
        for(int j=0;j<7;j++)
            total += arr[i][j];

    return total / 28.0;
}

int main() {
    int power[4][7];

    cout << "Enter data:\n";
    for(int i=0;i<4;i++)
        for(int j=0;j<7;j++)
            cin >> power[i][j];

    regionTotal(power);

    cout << "Highest Demand Day = " << highestDay(power)+1 << endl;
    cout << "Overall Average = " << overallAverage(power) << endl;

    return 0;
}
```

TASK 10 : CLASSROOM ATTENDANCE SYSTEM USING FUNTIONS.

```
#include <iostream>

using namespace std;


void studentAttendance(int arr[5][7]) {

    for(int i=0;i<5;i++) {

        int sum=0;

        for(int j=0;j<7;j++) sum += arr[i][j];

        cout << "Student " << i+1 << " Attendance = " << sum << endl;

    }

}


int highestAttendance(int arr[5][7]) {

    int maxStudent=0, maxDays=0;


    for(int i=0;i<5;i++) {

        int sum=0;

        for(int j=0;j<7;j++) sum += arr[i][j];


        if(sum > maxDays) {

            maxDays = sum;

            maxStudent = i;

        }

    }

    return maxStudent;

}
```

```
float overallPercentage(int arr[5][7]) {
    int total=0;
    for(int i=0;i<5;i++)
        for(int j=0;j<7;j++)
            total += arr[i][j];

    return (total / 35.0) * 100;
}

int main() {
    int att[5][7];

    cout << "Enter attendance (1=Present, 0=Absent):\n";
    for(int i=0;i<5;i++)
        for(int j=0;j<7;j++)
            cin >> att[i][j];

    studentAttendance(att);

    cout << "Highest Attendance Student = " << highestAttendance(att)+1 << endl;
    cout << "Overall Attendance Percentage = " << overallPercentage(att) << "%" << endl;

    return 0;
}
```